



Project Charter

Retail Business Intelligence Platform

Using the Olist Brazilian E-Commerce Dataset

Prepared by:

Moncef Amen Allah Ben Maatallah

Program:

Business Intelligence Engineering

Institution:

ESPRIT School of Engineering

Version: 1.0

Date: August 2026

Table of Contents

1. Project Overview
2. Project Background
3. Business Objectives
4. Project Scope
5. Out of Scope
6. Project Deliverables
7. Project Methodology
8. Project Timeline
9. Technologies and Tools
10. Stakeholders
11. Roles and Responsibilities
12. Risks and Constraints
13. Assumptions
14. Success Criteria
15. High-Level Architecture
16. Conclusion

1. Project Overview

1.1 Project Title

Retail Business Intelligence Platform

1.2 Purpose

The purpose of this project is to design and implement a complete Business Intelligence platform using the Brazilian Olist E-commerce dataset. The platform aims to transform raw transactional data into meaningful business insights through data warehousing, ETL processes, analytics, dashboards, machine learning, and workflow automation.

. 2. Project Background

Retail companies generate large volumes of transactional data every day.

However, this data is often distributed across multiple operational systems, making it difficult to perform comprehensive business analysis.

Without a centralized Business Intelligence platform, managers face challenges in monitoring sales performance, customer behavior, logistics efficiency, and overall business operations.

This project addresses these challenges by building a complete BI platform based on a real-world public dataset.

3. Business Objectives

The project aims to achieve the following objectives:

- Centralize business data into a unified Data Warehouse.
- Improve data quality through ETL processes.
- Enable interactive business reporting.
- Monitor key performance indicators (KPIs).
- Analyze customer behavior and sales performance.

- Evaluate seller and logistics performance.
- Apply machine learning techniques for predictive analytics.
- Automate reporting workflows using n8n.

4. Project Scope

The project includes:

- Business Analysis
 - Data Profiling
 - PostgreSQL Database Implementation
 - Staging Area Design
 - Data Warehouse Design
 - ETL Development using Talend
 - SQL Development
 - Power BI Dashboard Development
 - Exploratory Data Analysis
 - Machine Learning
 - Workflow Automation
 - Technical Documentation
 - GitHub Portfolio
-

5. Out of Scope

The following features are excluded:

- Online transaction processing (OLTP)
- Inventory management
- Payment gateway implementation
- Customer support systems
- Mobile application development
- Cloud deployment

6. Project Deliverables

The project will produce the following deliverables:

Deliverable	Description
Business Understanding	Business context and objectives
Project Charter	Project planning document
Business Requirements	Functional requirements
Source System Analysis	Analysis of source datasets
Data Profiling Report	Data quality assessment
PostgreSQL Database	Database implementation
Staging Area	Intermediate data storage
Data Warehouse	Dimensional data model
ETL Workflows	Talend ETL jobs
SQL Scripts	Database objects and queries
Power BI Dashboards	Interactive dashboards
Machine Learning Models	Predictive analytics
n8n Workflows	Automation processes
Technical Documentation	Complete project documentation

7. Project Methodology

The project follows a Business Intelligence development lifecycle inspired by CRISP-DM and Kimball methodologies.

The project phases include:

1. Business Understanding
2. Project Planning
3. Requirements Analysis
4. Source System Analysis
5. Data Profiling
6. Data Warehouse Design
7. ETL Development
8. Dashboard Development

- 9. Machine Learning
 - 10. Automation
 - 11. Documentation
 - 12. GitHub Publication
-

8. Project Timeline

Phase	Status
Business Understanding	Completed
Project Charter	In Progress
Business Requirements	Planned
Source System Analysis	Planned
Data Profiling	Planned
Database Design	Planned
ETL Development	Planned
Dashboard Development	Planned
Machine Learning	Planned
Automation	Planned
Documentation	Planned

9. Technologies and Tools

Category	Technology
Database	PostgreSQL
ETL	Talend Open Studio
Data Visualization	Power BI
Programming	Python
Query Language	SQL
Data Modeling	dbdiagram.io
Automation	n8n
Version Control	Git & GitHub
Documentation	Microsoft Word / PDF

10. Stakeholders

Stakeholder	Responsibility
CEO	Strategic decision-making
Sales Manager	Sales performance analysis
Marketing Manager	Customer analysis
Operations Manager	Logistics monitoring
Finance Manager	Financial performance
BI Engineer	Design and implementation of the BI platform
Project Developer	Moncef Ben Maatallah

11. Roles and Responsibilities

Role	Responsibilities
Project Manager	Project planning and coordination
Business Analyst	Business requirements analysis
BI Engineer	Data Warehouse design
ETL Developer	Data integration
Data Analyst	Dashboard development
ML Engineer	Predictive analytics
Technical Writer	Project documentation

12. Risks and Constraints

Risks

- Missing or incomplete data.
- Data quality issues.
- Limited historical data.
- Large dataset processing time.
- Learning curve for multiple technologies.

Constraints

- Single developer.
 - Academic project timeline.
 - Public dataset limitations.
 - No access to real-time operational data.
-

13. Assumptions

The project assumes that:

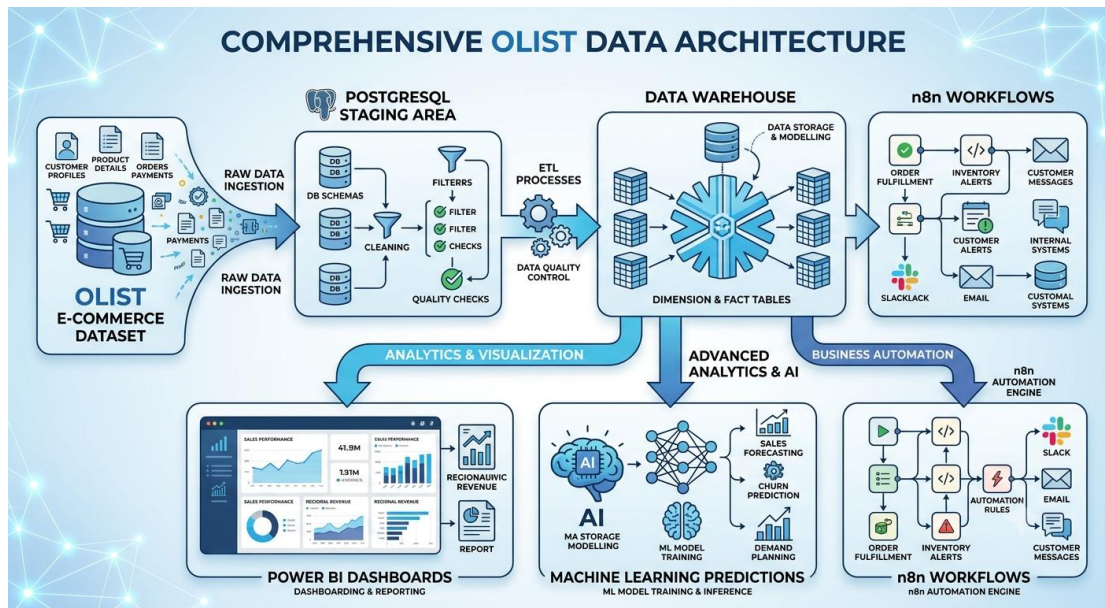
- The Olist dataset accurately represents retail business operations.
 - PostgreSQL is sufficient for the project scale.
 - Talend supports the required ETL workflows.
 - Power BI meets the reporting requirements.
 - The available data is adequate to answer the identified business questions.
-

14. Success Criteria

The project will be considered successful if:

- The Data Warehouse is successfully implemented.
- ETL workflows load data correctly.
- Dashboards answer business questions.
- KPIs are accurately calculated.
- Machine learning models produce meaningful results.
- Documentation is complete and well organized.
- The GitHub repository is structured professionally.

15. High-Level Architecture



16. Conclusion

This Project Charter defines the overall framework for the Retail Business Intelligence Platform project. It establishes the objectives, scope, methodology, deliverables, stakeholders, and success criteria that will guide the development of the solution.

By following a structured Business Intelligence methodology, this project aims to simulate a real enterprise implementation while demonstrating best practices in data engineering, analytics, reporting, and documentation.